

#### HARDWARE CORE / LOGISTICS-OPTIMIZED CARRIER

Single chassis with modular front/rear implements.



#### POSITIONING

## One carrier. Four-season utility.

- Multi-utility, multi-season chassis reduces idle equipment logic.
- Tracks and zero-turn maneuvering support narrow paths, slopes, and exclusion zones.
- Supervised autonomy keeps human override available for high-liability sites.

#### MODE

HUMAN-IN-LOOP

#### OUTPUT

GIS + LOGS

## 24 IN

**MAX SNOW DEPTH**

high-accumulation storms

## 40 DEG

**SLOPE CAPACITY**

52 deg peak noted

## 5-6 HR

**WINTER RUNTIME**

plow/deice or broom

## 1 MIN

**HOT-SWAP RECOVERY**

or wireless dock

#### MECHANICAL

##### Ruggedized carrier

- Approx. 1,500 lb empty; 2,000 lb loaded with implements.
- Standard U.S. pallet footprint for commercial-vehicle transport.

#### AUTONOMY + SAFETY

##### Compliance-first control

- LiDAR, Radar, visual cameras, RTK-GPS perception stack.
- 360-degree awareness, 6-ft monitor radius, remote override.

#### MOBILITY

##### Built for uneven outdoor work

- Industrial track propulsion for unstable terrain and winter surfaces.
- Zero-turn capability for hydrants, signage, walkways, and tight areas.

#### POWER + CHARGING

##### Recovery options for storm duty

- 150 Ah battery platform; task-specific runtime planning.
- Wireless autonomous docking through standard 110V outlet.

#### ATTACHMENTS

##### Single chassis, multi-task

- 4-6 ft straight/V-plows, power broom, mowing decks.
- Granular and liquid deicing systems for pre-treatment missions.

#### DATA LAYER

##### Operational record while work happens

- Geocoded images, cleared-area heatmaps, and field reports.
- ArcGIS/GeoJSON outputs and API options for customer systems.

## PLATFORM SPECIFICATION MATRIX

### Procurement-ready operating thresholds

|                |                                          |
|----------------|------------------------------------------|
| WEIGHT         | ~1,500 lb empty / ~2,000 lb loaded       |
| TRANSPORT      | Standard U.S. pallet footprint           |
| TERRAIN        | Tracked propulsion / zero-turn control   |
| SNOW           | Up to 24 in accumulation                 |
| SLOPE          | Up to 40 deg, 52 deg peak                |
| SPEED          | 2.0 mph autonomous / 5.0 mph manual      |
| POWER          | 150 Ah battery / 1-min swap / 110V dock  |
| RUNTIME        | 5–6 hr winter / multi-season utility     |
| OPERATING TEMP | -20°C to 40°C (-4°F to 104°F)            |
| INGRESS RATING | IP65 (weather and dust resistant)        |
| SAFETY         | 360° perception, e-stop, geofence, LIDAR |

## FLEET COMMAND APP / UI DIRECTION

UNIT\_04 ACTIVE

SLA ROUTE A-19



BATTERY

78%

UPTIME

98.4%

LOG 21:38 GEO-FENCE OK PHOTO AUDIT SAVED API QUEUED

## DATA + INTEGRITY LAYER

### Documented service evidence, not just equipment telemetry

#### BEFORE/AFTER MEDIA

Geotagged and timestamped images/videos

#### CLEARED-AREA HEATMAPS

Route completion and service coverage

#### GIS EXPORTS

ArcGIS / GeoJSON / API handoff

#### ESG LEDGER

Emissions and consumable-use tracking

## MODULAR MISSION MODES



#### SNOW

4–6 ft plow / V-blade / deicing support



#### BROOM

Active down-pressure for sidewalks, light snow, and zero-tolerance surfaces.



#### VEGETATION

Tracked access for hard-to-staff edges, corridors, and vegetation routes.



#### MOW

Finish mowing, rough cutting, and slope maintenance for outdoor assets.